

Hugo Gómez García

Computer Engineering & Mathematics Student | Artificial Intelligence / Machine Learning

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PROFILE

Double-degree student in Computer Engineering and Mathematics (five courses with Highest Honors) with a strong mathematical foundation geared toward *machine learning*. I build applied AI end to end: from high-performance C++ compute kernels and computer vision to model deployment and optimization, including *on-device* inference. First prize at Indra's AI ideas competition for ATM, and a finalist project at the Santander X university entrepreneurship awards.

EDUCATION

Double Bachelor's Degree in Computer Engineering and Mathematics 2024 - 2030
University of Oviedo Oviedo, Spain

Five courses awarded Highest Honors (*Matrícula de Honor* — Spain's top per-course grade). Relevant coursework: mathematical analysis, probability and statistics, computability, optimization and algorithms.

Microcredential in Quantum Computing 2026
University of Oviedo

Areas of interest: quantum key distribution (QKD), post-quantum cryptography, quantum machine learning (QML) and quantum optimization.

PROJECTS

MCMT — Real-time multi-camera multi-target tracking 1st prize • Indra (ATM)

- Two-tier architecture scaling to 1,300 simultaneous cameras: a five-gate geometric cascade resolves ~99% of associations, while a graph neural network (MPGNN) decides the ambiguous 1% off the critical path.
- C++20 kernels (extended Kalman filter, oriented 3D IoU, 2D→3D projection) bridged to Python via *nanobind* with zero-copy Apache Arrow memory; designed for free-threaded (GIL-free) Python with *lock-free* data structures.
- Visual re-identification (OSNet / CLIP) and a FAISS gallery for long-term re-association; validated on the WILDTRACK dataset with MODA/MODP metrics.

Tech: Python • C++20 • PyTorch • ONNX • FAISS • Apache Arrow • Eigen • NumPy

BioFit — AI health app • entrepreneurship project Finalist • Santander X

- Personalized virtual coach orchestrating training, nutrition and habits as an all-in-one health ecosystem.
- Computer-vision engine that analyzes exercise execution and corrects the user's form in real time.

Tech: Computer vision • Deep learning • Real-time AI

Operación Picos de Europa — Medical-imaging interoperability IABiomed 2026 Hackathon

- Pipeline connecting two real Spanish healthcare APIs (Idonia × Recog) to tackle portability of DICOM studies across regions and patient comprehension of radiology reports (Spanish Law 41/2002).
- Humanizes the report with an LLM and delivers it via *Magic Link*, but only after passing a four-layer validation belt (INFLEZ readability, clinical checklist, biomedical NER and an LLM semantic judge) that blocks delivery if clinical findings are omitted or downplayed.
- Python backend (FastAPI, Pydantic, interchangeable *stub/live* client architecture); reverse-engineered Idonia's undocumented JWT authentication.

Tech: Python • FastAPI • Pydantic • LLMs • Biomedical NER • REST / JWT APIs

EchoLens — On-device, in-browser audio classifier In progress GitHub

- Sound classifier running entirely in the browser, server-free: audio never leaves the user's device.
- Audio CNN trained in PyTorch (UrbanSound8K), exported to ONNX and quantized to INT8 (~4× smaller), run in real time with onnxruntime-web over WebAssembly; log-mel spectrograms computed live via the Web Audio API and an AudioWorklet.
- Combines audio DSP, deep learning, edge deployment and in-browser ML engineering, with CI/CD, tests and a quantitative latency/accuracy evaluation.

Tech: PyTorch • ONNX • onnxruntime-web • WebAssembly • Web Audio API • JavaScript

TECHNICAL SKILLS

Programming languages: Python, C++ (C++20), C, Java, JavaScript, SQL

AI & Machine Learning: PyTorch, deep learning (CNN, GNN), computer vision, LLMs, ONNX / ONNX Runtime, model quantization (INT8), FAISS

Scientific computing & data: NumPy, Apache Arrow, Eigen, signal processing (STFT, mel filters), nanobind (Python↔C++)

Backend & deployment: FastAPI, Pydantic, WebAssembly / onnxruntime-web, Web Audio API, SQLite, concurrency (free-threaded Python, *lock-free*)

Engineering & tooling: Git, CI/CD, pytest, CMake

LANGUAGES

Spanish (native) | English — fluent technical reading and conversation